

Residues at Simple Poles and Higher-Order Poles

Source: J. E. Santos, Cambridge Part IB Complex Methods, Example Sheet 2, Question 7.

1. Show that if $f(z)$ and $g(z)$ are analytic, and g has a simple zero at $z = z_0$, the residue of $f(z)/g(z)$ at $z = z_0$ is $f(z_0)/g'(z_0)$. In particular, show that $f(z)/(z - z_0)$ has residue $f(z_0)$.
2. Prove the formula for the residue of a function $f(z)$ that has a pole of order N at $z = z_0$:

$$\lim_{z \rightarrow z_0} \left\{ \frac{1}{(N-1)!} \frac{d^{N-1}}{dz^{N-1}} ((z - z_0)^N f(z)) \right\}. \quad (1.1)$$

Solution. The residue is given by

$$\text{Res} \left(\frac{f(z)}{g(z)}, z = z_0 \right) = \lim_{z \rightarrow z_0} \frac{(z - z_0)f(z)}{g(z)} = \lim_{z \rightarrow z_0} \frac{f(z) + (z - z_0)f'(z)}{g'(z)} = \frac{f(z_0)}{g'(z_0)}, \quad (1.2)$$

where we have used the fact that $g(z_0) \neq 0$ at the pole. Choosing $g(z) = z - z_0$, we find

$$\text{Res} \left(\frac{f(z)}{z - z_0}, z = z_0 \right) = f(z_0). \quad (1.3)$$

For a function $f(z)$ that has pole of order N at $z = z_0$, it admits a Laurent expansion of the form

$$f(z) = \sum_{n=1}^N \frac{b_n}{(z - z_0)^n} + \sum_{n=0}^{\infty} a_n (z - z_0)^n. \quad (1.4)$$

Multiplying $f(z)$ by $(z - z_0)^N$, we find

$$(z - z_0)^N f(z) = \sum_{n=1}^N b_n (z - z_0)^{N-n} + \sum_{n=0}^{\infty} a_n (z - z_0)^{N+n}. \quad (1.5)$$

It is therefore easy to see that

$$\text{Res} (f(z), z = z_0) = \lim_{z \rightarrow z_0} \left\{ \frac{1}{(N-1)!} \frac{d^{N-1}}{dz^{N-1}} ((z - z_0)^N f(z)) \right\}. \quad (1.6)$$